

LAGUERRE FUNCTIONS

Laguerre's Differential Equation

Laguerre ODE,

$$xy''(x) + (1-x)y'(x) + ny(x) = 0.$$

This ODE is not self-adjoint.

Rodrigues Formula

Letting $L_n(x)$ denote the n th Laguerre polynomial, the Rodrigues formula is (apart from a scale factor) given by:

$$L_n(x) = \frac{e^x}{n!} \left(\frac{d}{dx} \right)^n \left(x^n e^{-x} \right)$$

Generating Function

$$g(x, t) = \frac{e^x}{2\pi i} \oint_C \frac{e^{-z} dz}{z - x - tz}$$

Our integrand has a simple pole at $z = x/(1-t)$, with residue $e^{-x/(1-t)}/(1-t)$, and $g(x, t)$ reduces to

$$g(x, t) = \frac{e^{-xt}/(1-t)}{1-t} = \sum_{n=0}^{\infty} L_n(x) t^n$$

Recurrence Formula

$$(n+1)L_{n+1}(x) = (2n+1-x)L_n(x) - nL_{n-1}(x)$$

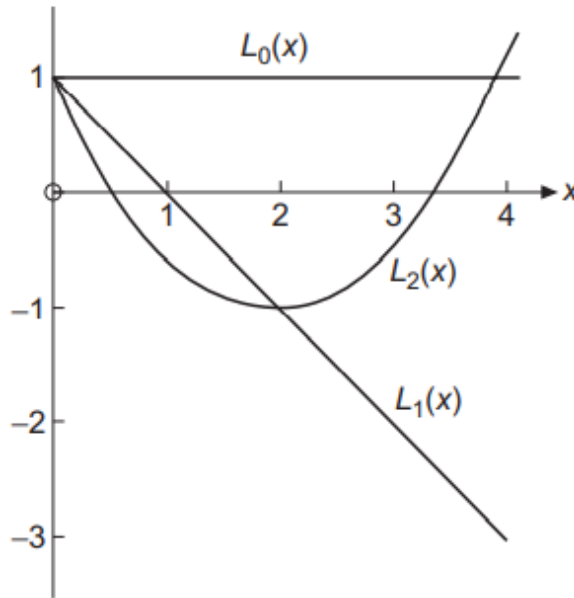
And

$$xL'_n(x) = nL_n(x) - nL_{n-1}(x).$$

To use these recurrence formulas we need starting values. From the Rodrigues formula, we easily find $L_0(x) = 1$ and $L_1(x) = 1 - x$. And by applying, we continue to $L_n(x)$ with $n > 1$,

obtaining the results given in below table:

$$\begin{aligned}
 L_0(x) &= 1 \\
 L_1(x) &= -x + 1 \\
 2!L_2(x) &= x^2 - 4x + 2 \\
 3!L_3(x) &= -x^3 + 9x^2 - 18x + 6 \\
 4!L_4(x) &= x^4 - 16x^3 + 72x^2 - 96x + 24 \\
 5!L_5(x) &= -x^5 + 25x^4 - 200x^3 + 600x^2 - 600x + 120 \\
 6!L_6(x) &= x^6 - 36x^5 + 450x^4 - 2400x^3 + 5400x^2 - 4320x + 720
 \end{aligned}$$



Laguerre – Polynomial

Orthogonality $L_n(x)$ must satisfy an orthogonality condition of the form

$$\int_0^{\infty} e^{-x} L_m(x) L_n(x) dx = \delta_{mn}$$

Example: Find the value of (a) $L'_n(0)$

(b) $L''_n(0)$

Solution:(a)

$$xL''_n + (1-x)$$

$$L'_n(x) + nL_n(x) = 0 \dots\dots (1)$$

$$L'_n(0) + nL_n(0) = 0$$

$$L'_n(0) = -nL_n(0) = -n$$

(b) Differentiate eqn. (1)

$$x L'''_n(x) + L''_n(x) + (1-x)L''_n(x) + L'_n(x)(-1) + L'_n(x) = 0$$

put $x = 0$

$$0 + L''_n(0) + L''_n(0) - L'_n(0) + n L'_n(0) = 0$$

$$2L''_n(0) = L'_n(0) - n L'_n(0)$$

$$L''_n(0) = \frac{1}{2} [-n + n^2] = \frac{n[n-1]}{2}$$

$$L''_n(0) = \frac{n(n-1)}{2}$$